Sensor Calibration Using AI

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***Abstract*—Sensor measurements are often affected by drift, noise, and nonlinear distortions, leading to deviations from the true physical quantity. Let the true measurand be denoted as x and the sensor output as $y = f(x)$. Calibration aims to recover x from y by learning an approximation of the inverse mapping $f^{-1}(y)$.**

In this project, artificial intelligence techniques are employed to correct drift in PM2.5 air quality sensors using the Seoul Air Pollution dataset. To simulate real-world conditions, drift is artificially introduced into the PM2.5 measurements. Several machine learning models are then trained to reconstruct the true signal from the corrupted data.

Four approaches are investigated: Linear Regression, Polynomial Regression, Random Forest Regression, and a Long Short-Term Memory (LSTM) network enhanced with residual learning and multi-feature input. Model performance is evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) across multiple test samples.

Experimental results demonstrate that all AI-based methods outperform a simple linear baseline in calibration accuracy, with the LSTM model achieving the best overall performance. These findings highlight the effectiveness of machine learning for sensor calibration and its potential to improve the reliability of real-world sensing systems.

***Index Terms*—Sensor calibration, drift, LSTM, Random Forest, PM2.5, regression.**

I. Introduction

Sensors are widely used to monitor physical phenomena such as air quality, temperature, and pressure. In an ideal setting, the sensor output y is a deterministic function of the true measurand x, expressed as $y = f(x)$. In practice, however, sensor characteristics evolve over time due to ageing, environmental influences, and hardware imperfections. These effects lead to sensor drift, where the measured value deviates from the true quantity even under constant conditions.

Sensor drift reduces the reliability of measurement systems and can negatively impact downstream applications, including environmental monitoring, health assessment, and industrial control. Calibration seeks to recover the true value x from the observed sensor output y by learning an approximation of the inverse mapping $f^{-1}(y)$.

Traditional calibration methods typically rely on manual adjustments, static correction curves, or simple linear models. While computationally efficient, these approaches often fail to capture complex and time-varying drift patterns. In contrast, machine learning (ML) and deep learning (DL) methods can learn nonlinear relationships directly from data, making them well-suited for modeling complex sensor behaviour and compensating for drift [2]. Making them well-suited for modeling sensor behavior and compensating for drift.

In this project, AI-based calibration algorithms are developed for PM2.5 air quality sensors using the Seoul Air Pollution dataset. To simulate realistic degradation, drift is artificially injected into PM2.5 measurements, and models are trained to reconstruct the true signal from corrupted observations. Four approaches are investigated: Linear Regression, Polynomial Regression, Random Forest Regression, and a Long Short-Term Memory (LSTM) network incorporating residual learning and multi-feature time-series inputs.

Model performance is evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) across multiple test samples. The results provide insight into the effectiveness of different ML approaches for sensor calibration and demonstrate the potential of AI-based methods to enhance the reliability of real-world sensing systems.

II. Requirements and Data Analysis

A. Project Requirements

The objective of this project is to design and evaluate AI algorithms capable of calibrating drift in PM2.5 sensor measurements. The system should learn to reconstruct the true PM2.5 value from a drifted sensor reading using supervised learning.

The main requirements are:

- The system must accept drifted PM2.5 measurements as input.
- The system must output calibrated PM2.5 values approximating the true signal.
- The system must use AI-based calibration algorithms.
- The system must be trained using paired drifted–true PM2.5 data.
- The system performance must be evaluated using RMSE and MAE.

B. System Specifications

Based on the project description and best practices in ML evaluation, the following measurable specifications are defined:

- **S1: Input Handling**— The system must process drifted PM2.5 data without errors.
- **S2: Output Consistency**— The calibrated output must have the same temporal resolution as the input.

- **S3: RMSE Improvement**— The best AI model must achieve a lower RMSE than the linear baseline.
- **S4: MAE Improvement**— The best AI model must achieve a lower MAE than the linear baseline.
- **S5: Stability Across Samples**— Performance must be stable across at least 500 test samples.
- **S6: Generalisation**— The gap between training and validation loss should remain small, indicating no severe overfitting.
- **S7: Evaluation Robustness**— Results must be reported using both RMSE and MAE.

C. Data Description

The dataset used in this project is the Seoul Air Pollution dataset, which contains hourly measurements of air pollutants and meteorological variables from multiple monitoring stations. For this project, PM2.5 is selected as the target variable for calibration.

The main steps are:

1 Load the raw dataset from the repository.
2 Clean the data by handling missing values and removing outliers.
3 Select relevant features such as PM2.5, PM10, NO2, O3, and CO.
4 Artificially inject drift into the PM2.5 signal to simulate sensor degradation.
5 Create paired data consisting of drifted PM2.5 (input) and true PM2.5 (target).

D. Data Visualisation

To understand the effect of drift and the calibration task, several visualisations are performed.

As shown in Fig. 4, the drifted signal deviates significantly from the true $PM_{2.5}$ values, while the LSTM-calibrated output closely follows the original signal.

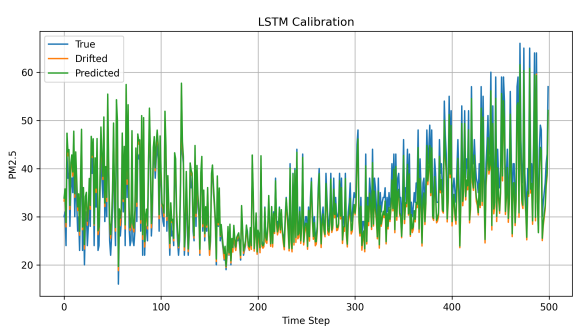

Fig. 1. Comparison of true $PM_{2.5}$ values, drifted sensor measurements, and LSTM-calibrated output over time. The figure illustrates how sensor drift introduces deviations from the true signal, while the LSTM model effectively reconstructs the original trend and reduces error.

The drifted signal deviates from the true PM2.5 values, especially over longer periods, while the calibrated output follows the true signal more closely.

An error distribution histogram is also generated to analyse the residuals between the calibrated output and the true PM2.5 values

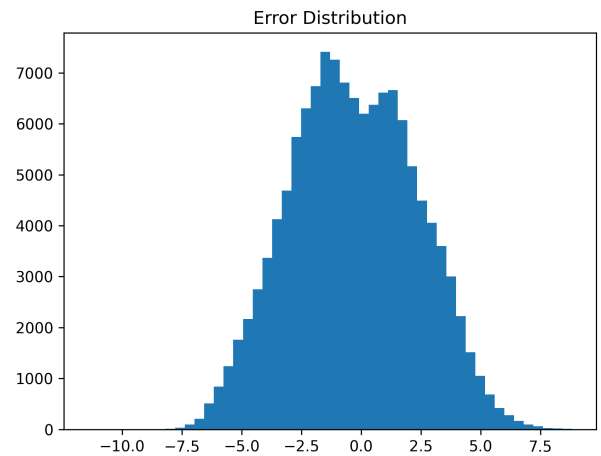

Fig. 2. Distribution of prediction errors for the LSTM calibration model. Most errors are concentrated around zero, indicating accurate calibration, with only a small number of larger deviations.

E. Literature Review

Machine learning techniques have been increasingly applied to sensor calibration and drift compensation problems. Traditional calibration methods, such as linear and polynomial regression, are simple and computationally efficient but often fail to capture complex non-linear relationships present in real-world sensor data.

Recent studies have shown that tree-based models and deep learning approaches significantly improve calibration accuracy. For example, [1] demonstrated that machine learning models can effectively model air quality data and compensate for noise and environmental variability. Similarly, [2] showed that advanced calibration techniques, including Random Forest and neural networks, can significantly reduce error in low-cost air quality sensors.

For time-series calibration tasks, recurrent neural networks such as Long Short-Term Memory (LSTM) networks are particularly effective due to their ability to capture temporal dependencies and long-term patterns in sequential data. This makes them well-suited for modeling sensor drift, which often evolves gradually over time.

Based on these findings, this project adopts a progressive approach: starting with simple regression models as baselines, followed by more advanced models such as Random Forest and LSTM to capture both non-linear relationships and temporal dependencies in the data.

F. Specification Justification

Reported results in the literature indicate that ML-based calibration can reduce RMSE and MAE compared to uncalibrated or linearly calibrated signals. Given the complexity of

air quality dynamics and sensor drift, it is realistic to expect that advanced models such as Random Forests and LSTMs will outperform simple linear baselines. The specifications S3 and S4 are therefore defined in terms of improvement over the linear model.

III. SYSTEM ENGINEERING

A. System Overview

Based on the specifications, a calibration system is designed to reconstruct true PM2.5 values from drifted sensor outputs. The system operates in two modes: training and evaluation. Figure 3 shows a block diagram of the proposed system.

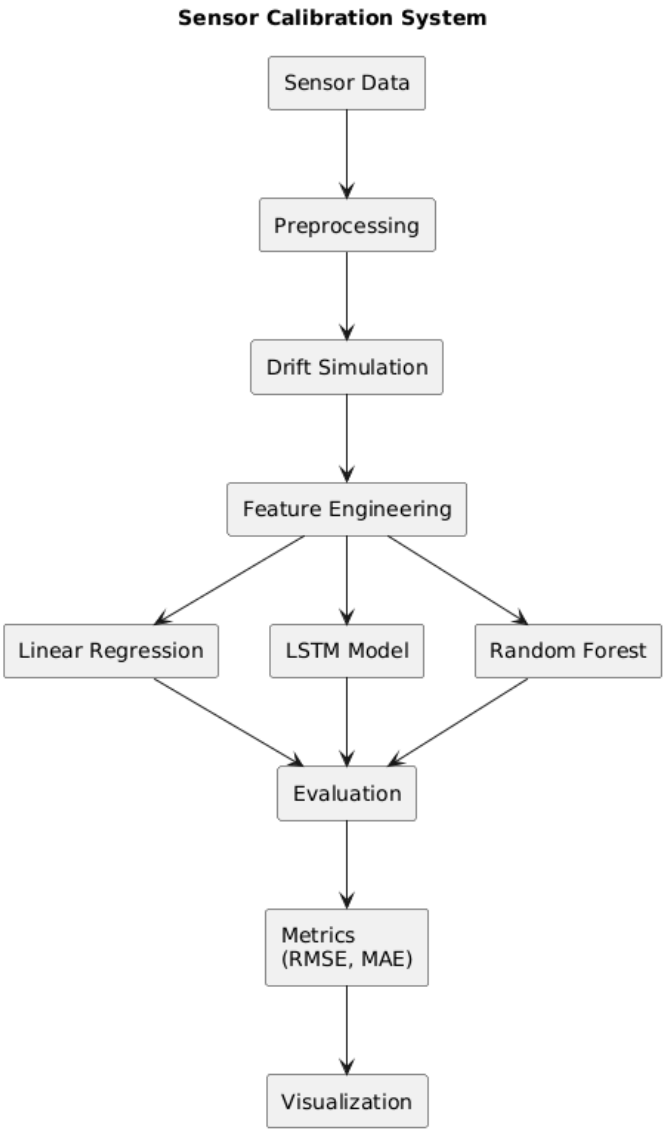

Fig. 3. Block diagram of the proposed AI-based sensor calibration system. The pipeline includes data preprocessing, drift injection, feature construction, model training, and evaluation, highlighting the transformation from drifted input to calibrated output.

High-level steps:

- Load and clean the dataset
- Inject drift into PM2.5 to create drifted measurements.
- For LSTM, construct sequences of multi-feature inputs.
- Train calibration models using drifted input and true target.
- Evaluate models on a held-out test set using RMSE and MAE.

B. Acceptance Test Procedures (ATP)

Table I summarises the ATPs used to verify that the system meets the specifications.

TABLE I
ACCEPTANCE TEST PROCEDURES FOR SYSTEM VERIFICATION

Test Description	Input	Expected Output
Input validation	Drifted PM2.5 data	System processes data without errors
Calibration accuracy	Drifted + true PM2.5	RMSE lower than linear baseline
Model comparison	Test set	LSTM achieves best RMSE/MAE
Error stability	500+ test samples	Error distribution without extreme outliers
Sequence handling	Time-series input	Correct sequence construction for LSTM

IV. ALGORITHM DESIGN

A. Overview of Approach

The algorithm design follows an iterative approach. This approach aligns with engineering design methodologies where simple baseline models are first evaluated before introducing more complex architectures to improve performance. In the first iteration, simple baseline models are implemented and evaluated. In the second iteration, more advanced models are introduced to improve performance.

B. Iteration 1 Baseline Models

In Iteration 1, two baseline models are implemented:

- **Linear Regression**: A simple linear model mapping drifted $PM_{2.5}$ measurements to true $PM_{2.5}$ values.
- **Random Forest Regression** A tree-based ensemble model using multi-feature input.

The dataset is split into training and test sets using an 80/20 split. Models are trained on the training set and evaluated on the test set. Table II shows the results of Iteration 1.

Motivation for Model Selection:

The choice of Linear Regression and Random Forest is motivated by the need to establish both a simple baseline and a more advanced non-linear reference model:

- **Baseline Comparison (Linear Regression):** Linear Regression is selected as a fundamental baseline due to its simplicity, interpretability, and low computational cost. It provides a reference point to evaluate whether more complex models offer meaningful improvements.
- **Capturing Non-Linear Relationships (Random Forest):** Sensor drift and environmental data often exhibit non-linear patterns. Random Forest is chosen because it can model complex, non-linear relationships without requiring explicit feature transformations.
- **Robustness to Noise and Overfitting:** Random Forest is inherently robust to noise and reduces overfitting through ensemble averaging. This makes it suitable for

sensor data, which may contain measurement errors and variability.

- **Benchmark for Model Progression:** Together, these two models provide a progression from simple to more sophisticated approaches, forming a strong foundation for later comparison with temporal models such as LSTM.

TABLE II
ITERATION 1 MODEL PERFORMANCE

Model	RMSE	MAE
Linear Regression	3.48	3.09
Random Forest	2.95	2.51

The Random Forest model improves over the linear baseline but still leaves significant residual error, motivating the use of more powerful models.

C. Improvement Strategy

Based on the limitations observed in Iteration 1, the following improvements are proposed:

- Use additional features (PM10, NO2, O3, CO) as inputs.
- Model temporal dependencies using sequence-based models.
- Employ non-linear models such as Random Forests and LSTMs.
- Use residual learning, where the model learns the correction between drifted and true values.

D. Iteration 2: Advanced Models

In Iteration 2, an advanced model was implemented:

- **LSTM Calibration Model** A recurrent neural network that takes sequences of multi-feature inputs and predicts the residual correction.

For the LSTM model, the input features include drifted PM2.5, PM10, NO2, O3, and CO. A sequence length of 5 time steps is used. Residual learning is applied by defining the target as the difference between true and drifted PM2.5. After prediction, the residual is added back to the drifted signal to obtain the calibrated output.

Motivation for LSTM:

The selection of the LSTM model is motivated by the limitations observed in Iteration 1 and the temporal nature of sensor drift:

- **Temporal Dependency Modeling:** Sensor drift evolves over time and cannot be fully captured by static models. LSTM networks are specifically designed to model sequential data and learn temporal dependencies, making them well-suited for this task.
- **Handling Time-Dependent Drift Patterns:** Unlike Linear Regression and Random Forest, LSTM can capture long-term dependencies and gradual changes in sensor behavior, which is critical for correcting drift that accumulates over time.
- **Improved Performance for Sequential Data:** LSTM has been widely used in time-series prediction problems due to its ability to retain relevant past information through memory cells. This allows the model to better generalize under different drift conditions.
- **Residual Learning Capability:** By predicting the correction (residual) instead of the absolute value, the LSTM model focuses on learning the error pattern introduced by drift, which simplifies the learning task and improves accuracy.
- **Extension of Spiral Approach:** Following the spiral development methodology, LSTM represents a natural progression from static and ensemble models to a temporal deep learning approach that addresses previously identified limitations.

E. Training Analysis

The LSTM model is trained for 25 epochs using the Adam optimizer and MSE loss. Training and validation losses are recorded.

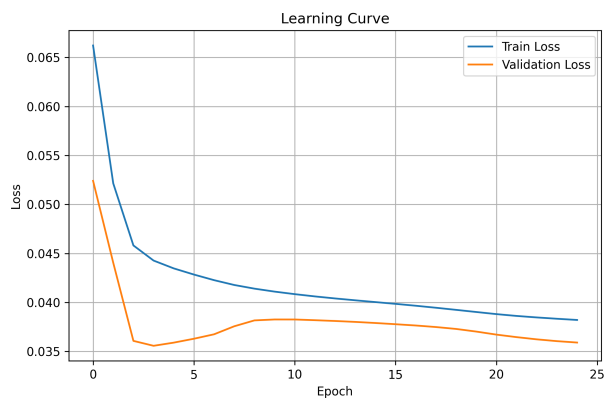

Fig. 4. Training and validation loss curves for the LSTM calibration model across epochs. Both curves decrease consistently and remain close, indicating stable learning and good generalisation without significant overfitting.

The learning curves show a steady decrease in both training and validation loss, with no significant divergence, indicating good generalisation.

F. Iteration 2 Results

Table III summarises the performance of the advanced models.

TABLE III
ITERATION 2 MODEL PERFORMANCE

Model	RMSE	MAE
Linear Regression	3.48	3.09
Random Forest	2.95	2.51
LSTM	2.66	2.19

The LSTM model achieves the lowest RMSE and MAE, satisfying the specifications S3 and S4 and demonstrating the benefit of sequence modeling and residual learning.

G. Visual Comparison of Calibration Methods

To better understand the calibration quality, a visual comparison is performed by plotting the true PM2.5 signal, the drifted sensor output, and the LSTM-calibrated output over a selected time window, as shown in Figure 1.

The drifted signal shows a clear offset and distortion compared to the true signal, while the calibrated output closely follows the true PM2.5 values, especially around peaks and transitions.

H. Error Analysis

The error distribution of the LSTM model is shown in Figure 2. Most errors are concentrated around zero, with relatively few large deviations, indicating stable calibration performance.

V. CONSOLIDATION OF ATPS

Table IV summarises whether each specification has been met.

TABLE IV
SPECIFICATION VERIFICATION

ID	Description	Met?
S1	Input Handling	Yes
S2	Output Consistency	Yes
S3	RMSE Improvement	Yes (LSTM $<$ Linear)
S4	MAE Improvement	Yes (LSTM $<$ Linear)
S5	Stability Across Samples	Yes (500+ test samples)
S6	Generalisation	Yes (no strong overfitting)
S7	Evaluation Robustness	Yes (RMSE and MAE reported)

VI. CONCLUSION

This project investigated AI-based methods for sensor calibration using the Seoul Air Pollution dataset. Drift was artificially injected into PM2.5 measurements to simulate real-world sensor degradation, and multiple models were trained to reconstruct the true signal from corrupted inputs.

Baseline models such as Linear Regression provided a simple reference but were limited in handling non-linear and temporal patterns. More advanced models, including Random Forest and LSTM, demonstrated improved performance. In particular, the LSTM model achieved the lowest RMSE and MAE, highlighting the importance of temporal modeling and residual learning in calibration tasks.

The results confirm that AI-based calibration methods can significantly improve sensor accuracy and robustness. This is especially important for applications such as air quality monitoring, where reliable measurements are critical.

Future work could explore transformer-based models, uncertainty-aware calibration, and deployment on real-time embedded systems for practical applications.

The experimental results also demonstrate that incorporating temporal dependencies significantly improves calibration performance compared to static models.

VII. ADMIN PART

A. GitHub Repository

A placeholder repository for this project is: https://github.com/iqra-rafique-swed/Sensor-Callibration-using-AI

B. EU AI Act Impact

The developed system performs numerical regression for sensor calibration and does not directly make safety-critical decisions. It can be considered a low-risk AI system under the EU AI Act, provided it is used as a decision-support tool rather than an autonomous controller.

C. Work Distribution

All work in this project, including data processing, model implementation, training, evaluation, and report writing, was carried out individually.

D. AI Usage

AI tools were used for code debugging assistance and for improving the structure and clarity of the written report. No AI system was used to generate experimental results.

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